

Lecture-7

Mathematics 1 (15B11MA111)

Module: Partial Differentiation

CO1 [C105.1]

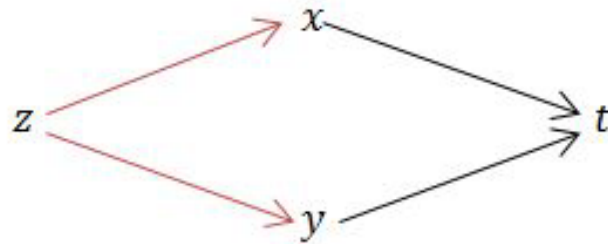
Topics to be covered

- **Derivative of composite and implicit functions (Chain Rule)**
- **Change of Variables**
- **Reference for the lecture**
R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Derivative of composite and implicit functions

➤ Chain Rule I:

Let $z = f(x, y)$ be a function of two independent variables x and y and x, y are also the function of some t such that as, $x = \phi(t), y = \varphi(t)$



$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \quad (1)$$

Derivative of composite and implicit functions

➤ Note:

1. If $z = f(x_1, x_2, \dots, x_n)$ and $x_i = \phi_i(t)$, then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x_1} \cdot \frac{dx_1}{dt} + \dots + \frac{\partial z}{\partial x_n} \cdot \frac{dx_n}{dt}$$

2. If $z = f(x, y)$ and $y = \phi(x)$, then

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx}$$

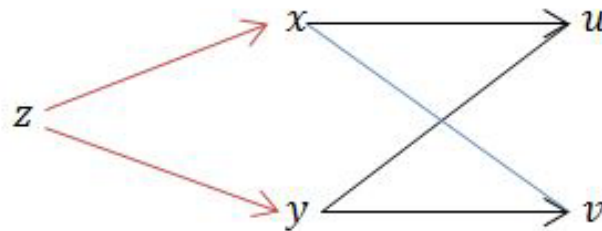
3. For Implicit function $f(x, y) = c$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

Derivative of composite and implicit functions Contd...

➤ Chain Rule II:

Again if x, y are functions of u, v as $x = \phi(u, v), y = \varphi(u, v)$ then



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \quad (2)$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \quad (3)$$

Derivative of composite and implicit functions Contd...

➤ Example 1:

Find $\frac{df}{dt}$ at $t = 0$, where

$$(i) \ f(x, y) = x \cos y + e^x \sin y$$
$$x = t^2 + 1; y = t^3 + t.$$

$$(ii) \ f(x, y, z) = x^3 + xz^3 + y^3 + xyz$$
$$x = e^t, y = \cos t, z = t^3$$

Derivative of composite and implicit functions Contd...

➤ Solution

(i)

$$\frac{df}{dt} = (\cos y + e^x \sin y)(2t) + (-x \sin y + e^x \cos y)(3t^2 + 1)$$

$$\frac{df}{dt} = e$$

(Since at $t = 0$, $x = 1$, $y = 0$)

(ii) $\frac{df}{dt} = 3.$

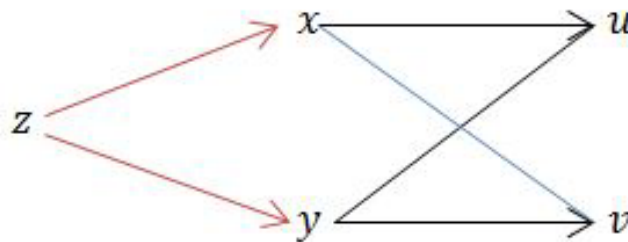
Derivative of composite and implicit functions Contd...

➤ Example 2:

If $z = f(x, y)$, $x = e^{2u} + e^{-2v}$ and $y = e^{-2u} + e^{2v}$, then show that

$$\frac{\partial f}{\partial u} - \frac{\partial f}{\partial v} = 2 \left(x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} \right).$$

Solution:



$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}$$

Derivative of composite and implicit functions Contd...

Solution contd:

$$\begin{aligned}\Rightarrow \frac{\partial f}{\partial u} &= \frac{\partial f}{\partial x} \cdot 2e^{2u} + (-2)e^{-2u} \cdot \frac{\partial f}{\partial y} \\ \Rightarrow \frac{\partial f}{\partial u} &= 2 \left(\frac{\partial f}{\partial x} \cdot e^{2u} - e^{-2u} \cdot \frac{\partial f}{\partial y} \right) \quad (4)\end{aligned}$$

And

$$\begin{aligned}\frac{\partial f}{\partial v} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v} \\ \Rightarrow \frac{\partial f}{\partial v} &= \frac{\partial f}{\partial x} \cdot (-2)e^{2v} + (2)e^{-2v} \cdot \frac{\partial f}{\partial y} \\ \Rightarrow \frac{\partial f}{\partial v} &= 2 \left(\frac{\partial f}{\partial y} \cdot e^{2v} - e^{-2v} \cdot \frac{\partial f}{\partial x} \right) \quad (5)\end{aligned}$$

Derivative of composite and implicit functions Contd...

Solution contd:

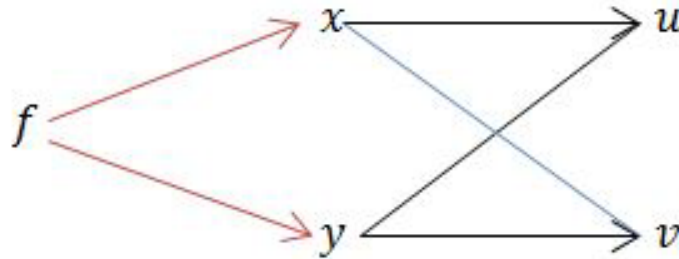
From eqns (4) and (5),

$$\begin{aligned}\frac{\partial f}{\partial u} - \frac{\partial f}{\partial v} &= 2 \left(\frac{\partial f}{\partial x} (e^{2u} + e^{-2v}) - \frac{\partial f}{\partial y} (e^{-2u} + e^{2v}) \right) \\ &= 2 \left(x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} \right)\end{aligned}$$

Change of Variables

➤ For two variables:

Let $f(x, y)$ be a function of two independent variables x and y and x, y are also the function of some u, v such that as, $x = \phi(u, v), y = \varphi(u, v)$



$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u} \quad (6)$$

To evaluate $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ in terms of $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$, we use Cramer's rule.

Change of Variables

➤ For two variables Contd:

For,

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Consider the determinants

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}, D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

$$\text{Then, } x = \frac{D_x}{D}, y = \frac{D_y}{D}.$$

Writing the system of eqns (6) as

$$\begin{aligned} \frac{\partial f}{\partial u} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \frac{\partial f}{\partial v} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v} \end{aligned}$$

Change of Variables

➤ For two variables Contd:

Therefore,

$$\frac{\partial f}{\partial x} = \frac{\begin{vmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \\ \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \end{vmatrix}}{\begin{vmatrix} \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \end{vmatrix}} = \frac{\frac{\partial(f, y)}{\partial(u, v)}}{J}$$

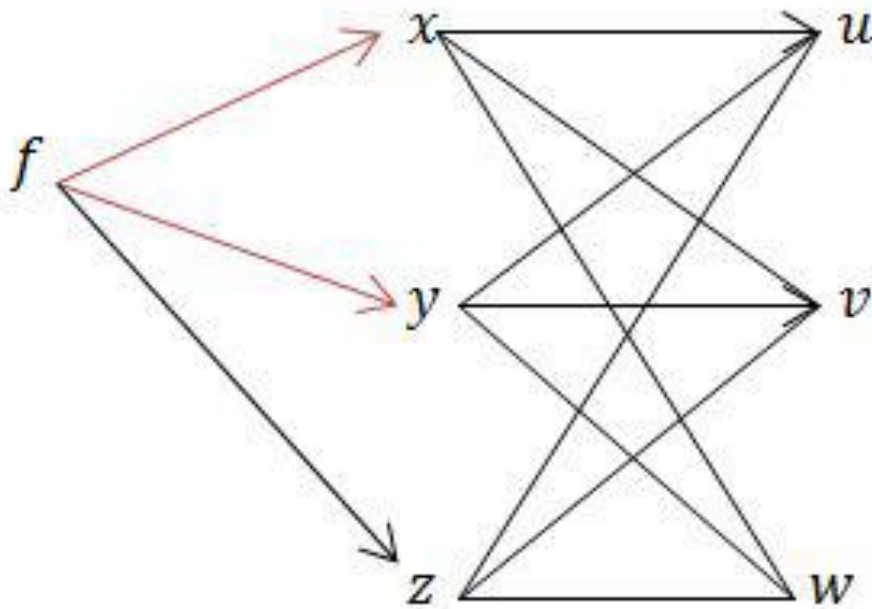
Similarly

$$\frac{\partial f}{\partial y} = -\frac{\frac{\partial(f, x)}{\partial(u, v)}}{J}$$

$$\text{where } J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

Change of Variables

➤ For Three Variables



Change of Variables

➤ For Three Variables Contd:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{1}{J} \frac{\partial(f, y, z)}{\partial(u, v, w)} \\ \frac{\partial f}{\partial y} &= -\frac{1}{J} \frac{\partial(f, x, z)}{\partial(u, v, w)} \\ \frac{\partial f}{\partial z} &= \frac{1}{J} \frac{\partial(f, x, y)}{\partial(u, v, w)}\end{aligned}$$

And

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix}$$

This J is called the Jacobian of x, y, z with respect to u, v, w .

Change of Variables

➤ Example 3:

If $z = f(x, y)$, $x = r \cos\theta$ and $y = r \sin\theta$, then show that

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = \left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial f}{\partial \theta}\right)^2.$$

Solution:

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$
$$\Rightarrow J = \begin{vmatrix} \cos\theta & \sin\theta \\ -r \sin\theta & r \cos\theta \end{vmatrix} = r$$

Change of Variables

Solution contd:

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r}$$
$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

To applying the Cramer's rule for the above expressions, we get

$$\frac{\partial f}{\partial x} = \frac{\begin{vmatrix} \frac{\partial f}{\partial r} & \sin\theta \\ \frac{\partial f}{\partial \theta} & r \cos\theta \end{vmatrix}}{J} = \frac{\partial f}{\partial r} \cos\theta - \frac{\sin\theta}{r} \frac{\partial f}{\partial \theta} \quad (7)$$

Change of Variables

Solution contd: and

$$\frac{\partial f}{\partial y} = \frac{\begin{vmatrix} \cos\theta & \frac{\partial f}{\partial r} \\ -r \sin\theta & \frac{\partial f}{\partial \theta} \end{vmatrix}}{J} = \sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r} \frac{\partial f}{\partial \theta} \quad (8)$$

Squaring and adding eqns (7) and (8), we obtain

$$\begin{aligned} & \left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \\ &= \left(\frac{\partial f}{\partial r} \cos\theta - \frac{\sin\theta}{r} \frac{\partial f}{\partial \theta} \right)^2 + \left(\sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r} \frac{\partial f}{\partial \theta} \right)^2 \\ &= \left(\frac{\partial f}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial f}{\partial \theta} \right)^2 \end{aligned}$$

THANK YOU

